Chapter 9: Gases & Kinetic Molecular Theory

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PHY201

Lecture 9

Part I

Gas Pressure

Outline of Part I: Gas Pressure

Outline of Part I: Gas Pressure

Introduction to Gases

Gas Pressure

Properties of Gases

What Makes Gases Unique?

- Gases **expand to fill** any container they occupy.
- Gases are highly **compressible**: volume changes substantially with pressure.
- Gases have much **lower density** than solids or liquids.
- Gases and liquids are both **fluids**: they flow and take the shape of their container.

- Gases **are** compressible.
- Liquids are **not** compressible.

Outline of Part I: Gas Pressure

Introduction to Gases

Gas Pressure

Pressure

Definition: Pressure (P) — SI Unit: pascal (Pa)

Pressure is the force exerted per unit area on a surface:

$$P = \frac{F}{A}$$

where F is force (N) and A is area (m^2).

- **Atmospheric pressure** arises from the weight of the air above Earth's surface.
- **Gas pressure** in a container results from molecules colliding with the container walls.

Units of Pressure

Unit	Symbol	Value at Standard Conditions
Pascal	Pa	101 325 Pa
Kilopascal	kPa	101.325 kPa
Atmosphere	atm	1 atm (definition)
Bar	bar	$\approx$ 1.013 25 bar
Millimeters of Hg	mmHg	760 mmHg
Torr	torr	760 torr
Pounds per sq. in.	psi	14.696 psi

- 1 mmHg $\approx$ 1 torr (essentially identical)
- Conversions: 1 atm = 760 torr = 101 325 Pa

Check Your Understanding

Convert 1.50 atm to (a) torr and (b) kPa.

Come to lecture to see the answer.

Manometers

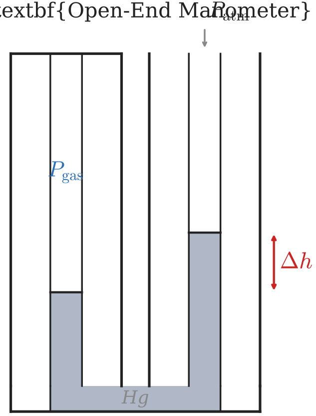
Measuring Gas Pressure

A **manometer** is a U-shaped tube containing mercury used to measure the pressure of a trapped gas.

Open-end manometer:

- One arm connects to the gas sample.
- The other arm is open to the atmosphere.
- The mercury level difference Δh gives the pressure difference.

\textbf{Open-End Manometer}



$$P_{\text{gas}} = P_{\text{atm}} \pm \Delta h \text{ (in mmHg)}$$

Reading a Manometer

Example

$P_{\text{atm}} = 760\,\text{mmHg}$. The gas side of the manometer is 30 mm **lower** than the open side. What is P_{gas}?

$$P_{\text{gas}} = P_{\text{atm}} + \Delta h$$

$$P_{\text{gas}} = 760\,\text{mmHg} + 30\,\text{mmHg}$$

$$\boxed{P_{\text{gas}} = 790\,\text{mmHg}}$$

- Gas side *lower* $\Rightarrow$ gas pressure is *greater* than atmospheric.
- Gas side *higher* $\Rightarrow$ gas pressure is *less* than atmospheric.

Check Your Understanding

A gas sample is connected to an open-end manometer.

$P_{\text{atm}} = 760\,\text{torr}$.

The mercury on the **gas side is** 45 mm **higher** than the open side.

What is P_{gas}?

Come to lecture to see the answer.

Part II

Gas Laws

Outline of Part II: Gas Laws

Outline of Part II: Gas Laws

Variables That Describe a Gas

The state of any gas sample is fully described by four variables:

Variable	Symbol	SI Unit	Common Units
Pressure	P	Pa	atm, mmHg, torr, kPa
Volume	V	m^3	L, mL
Temperature	T	K	Must convert from °C
Amount	n	mol	—

Temperature **must always be in Kelvin** in gas law calculations.

$T(\mathrm{K}) = T(^\circ C) + 273.15$

Temperature Scales

Converting Between Temperature Scales

$$T_K = T_C + 273.15$$

$$T_C = \tfrac{5}{9}(T_F - 32)$$

$$T_F = \tfrac{9}{5}\,T_C + 32$$

Example

Convert 25.0 °C to Kelvin.

$$T_K = 25.0\,°\mathrm{C} + 273.15 = \boxed{298.15\,\mathrm{K}}$$

Outline of Part II: Gas Laws

Boyle's Law

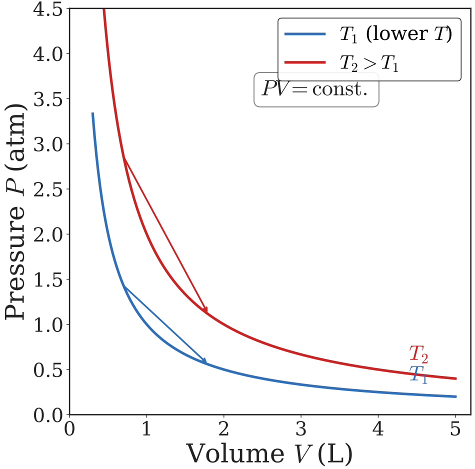
Boyle's Law (constant T, n)

At constant temperature and amount, pressure and volume are **inversely proportional**:

$$V \propto \frac{1}{P} \quad \Rightarrow \quad PV = \text{const.}$$

$$P_1 V_1 = P_2 V_2$$

Squeezing a sealed balloon: volume $\downarrow \Rightarrow$ pressure $\uparrow$.



Check Your Understanding

A gas occupies 4.0 L at 1.0 atm.

The pressure is increased to 2.0 atm at constant temperature.

What is the new volume?

Come to lecture to see the answer.

Charles's Law

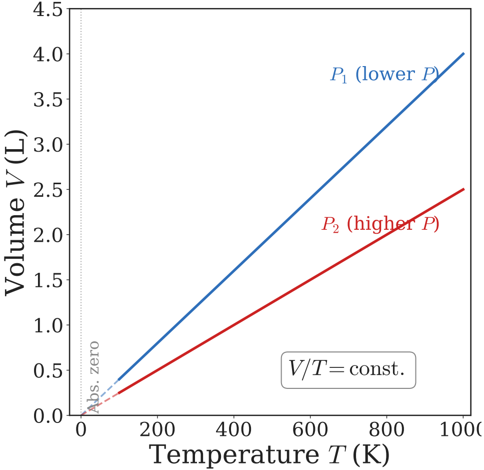
Charles's Law (constant P, n)

At constant pressure and amount, volume is **directly proportional** to temperature (in K):

$$V \propto T \quad \Rightarrow \quad \frac{V}{T} = \text{const.}$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

A balloon placed in a freezer shrinks; in a hot car, it expands.



Check Your Understanding

A balloon holds 2.0 L at 300 K.

It is cooled to 150 K at constant pressure.

What is the new volume?

Come to lecture to see the answer.

Avogadro's Law

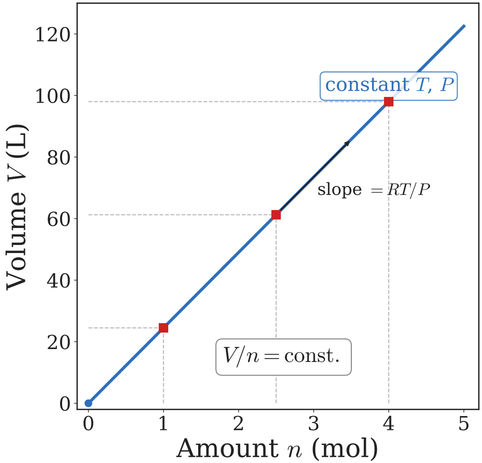
Avogadro's Law (constant T, P)

At constant temperature and pressure, volume is **directly proportional** to the amount of gas:

$$V \propto n \quad \Rightarrow \quad \frac{V}{n} = \text{const.}$$

$$\frac{V_1}{n_1} = \frac{V_2}{n_2}$$

A punctured tire: gas escapes (moles $\downarrow$) so volume $\downarrow$.



Summary: The Simple Gas Laws

Name	**Relationship**	**Held Const.**	**Example**
Boyle's	$V \propto 1/P$	T, n	Squeezing a balloon
Charles's	$V \propto T$	P, n	Balloon in freezer
Avogadro's	$V \propto n$	T, P	Punctured tire

- Gases are **compressible**: volume responds to pressure changes.
- Liquids are **incompressible**: volume does not change with pressure.

Check Your Understanding

A gas sample occupies 8.0 L and contains 2.0 mol at constant T and P.

If gas is added until the amount reaches 3.0 mol, what is the new volume?

Come to lecture to see the answer.

Outline of Part II: Gas Laws

Combining the Gas Laws: Ideal Gas Law

Combining all three simple gas laws:

$$V \propto \frac{1}{P}, \quad V \propto T, \quad V \propto n \quad \Longrightarrow \quad V \propto \frac{nT}{P}$$

Inserting a proportionality constant R (the **gas constant**):

$$PV = nRT$$

$R = 0.08206\ \mathrm{L{\cdot}atm{\cdot}mol^{-1}{\cdot}K^{-1}}$ (use when P in atm, V in L)

$R = 8.314\ \mathrm{J{\cdot}mol^{-1}{\cdot}K^{-1}}$ (SI; when P in Pa, V in $\mathrm{m^3}$)

Ideal Behavior and Conditions

- The ideal gas law assumes gas molecules have **no volume** and **no intermolecular forces**.
- Gases behave most **ideally** at high temperature and low pressure.

$$\text{Ideality} \propto T$$

Higher T: kinetic energy dominates over intermolecular attractions

$$\text{Ideality} \propto \frac{1}{P}$$

Lower P: molecules are far apart and rarely interact

Standard Temperature and Pressure (STP)

Standard Temperature and Pressure (STP)

- $T = 0\,°\text{C} = 273.15\,\text{K}$
- $P = 1\,\text{atm} = 101\,325\,\text{Pa}$

Standard Molar Volume

One mole of any *ideal* gas at STP occupies:

$$V = \frac{nRT}{P} = \frac{(1\,\text{mol})(0.08206\ \text{L}\cdot\text{atm}\cdot\text{mol}^{-1}\cdot\text{K}^{-1})(273.15\,\text{K})}{1\,\text{atm}} = \boxed{22.41\,\text{L}}$$

$22.41\,\text{L}\,\text{mol}^{-1}$ is the **standard molar volume** — a useful shortcut for STP problems.

Example: Using the Ideal Gas Law

Scuba Tank (cp2e_ch9)

A scuba tank contains 13.2 L of air at 153 atm.
What volume of air at 1.00 atm and 25.0 °C does this represent?
(Use combined gas law, n constant.)

$$P_1 = 153\,\text{atm}, \quad V_1 = 13.2\,\text{L}, \quad T_1 = 298\,\text{K}$$
$$P_2 = 1.00\,\text{atm}, \quad V_2 = ?, \quad T_2 = 298\,\text{K}$$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$V_2 = \frac{P_1 V_1 T_2}{T_1 P_2} = \frac{(153\,\text{atm})(13.2\,\text{L})(298\,\text{K})}{(298\,\text{K})(1.00\,\text{atm})}$$

$$\boxed{V_2 = 2020\,\text{L}}$$

- $T_1 = T_2$, so the temperature ratio cancels.
- 2020 L at atmospheric pressure — a huge volume for diving!

Relating Two States: The Combined Gas Law

When the amount of gas is constant (n fixed), PV/T is constant:

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

Special cases (one variable held fixed):

- T constant: $P_1 V_1 = P_2 V_2$ (Boyle's Law)
- P constant: $V_1/T_1 = V_2/T_2$ (Charles's Law)
- V constant: $P_1/T_1 = P_2/T_2$ (Amontons's Law)

Example: Relating Two States

State 1: $P_1 = 1.00\,\text{atm}$, $T_1 = 25.00\,°\text{C}$, $V_1 = 1.00\,\text{L}$.
Cool to $T_2 = -5.00\,°\text{C}$ at constant pressure. Find V_2.

$$T_1 = 298.15\,\text{K}, \quad T_2 = 268.15\,\text{K}$$

$$P_1 = P_2 \text{ (constant)}$$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$V_2 = V_1 \cdot \frac{T_2}{T_1} \cdot \underbrace{\frac{P_1}{P_2}}_{=1}$$

$$V_2 = 1.00\,\text{L} \cdot \frac{268.15\,\text{K}}{298.15\,\text{K}}$$

$$\boxed{V_2 = 0.899\,\text{L}}$$

- Always convert T to Kelvin first!
- Cooling $\Rightarrow$ volume decreases (Charles's Law)

Density and Molar Mass from the Ideal Gas Law

Density of an Ideal Gas

From $PV = nRT$ and $n = m/\mathcal{M}$:

$$\rho = \frac{m}{V} = \frac{P\mathcal{M}}{RT}$$

where $\mathcal{M}$ is molar mass.

Higher P or lower $T \Rightarrow$ denser gas.

Molar Mass of an Unknown Gas

Rearranging $PV = \frac{m}{\mathcal{M}} RT$:

$$\mathcal{M} = \frac{mRT}{PV}$$

Measure m, P, V, T to identify an unknown gas.

- These formulas let you use a gas as if it were a solid sample: measure its mass and gas properties, then find its molar mass.

Check Your Understanding

At the same temperature and pressure, which gas is **densest**?

(A) He (B) H_2 (C) N_2 (D) CO_2

Come to lecture to see the answer.

Part III

Gas Mixtures

Outline of Part III: Gas Mixtures

Dalton's Law and Partial Pressures

Gas Stoichiometry

Outline of Part III: Gas Mixtures

Mixtures of Gases: Dalton's Law

Dalton's Law of Partial Pressures

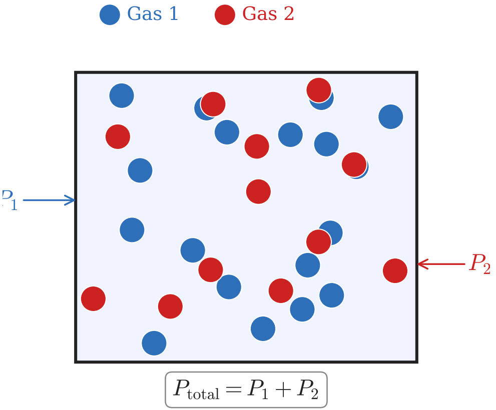
Each gas in a mixture exerts pressure **independently** of the others.

The **total pressure** equals the sum of all **partial pressures**:

$$P_{\text{total}} = \sum_i P_i = P_1 + P_2 + P_3 + \cdots$$

Dry Air at $760\,\text{mmHg}$

$P_{N_2} \approx 593,\ P_{O_2} \approx 160,\ P_{Ar} \approx 7$ mmHg
$P_{\text{total}} \approx 760\,\text{mmHg}$ ✓



Mole Fraction and Partial Pressure

Definition: Mole Fraction (χ_i)

The mole fraction of component i is the fraction of total moles it contributes:

$$\chi_i = \frac{n_i}{n_{\text{total}}}, \qquad \sum_i \chi_i = 1$$

Partial Pressure from Mole Fraction

$$P_i = \chi_i \times P_{\text{total}}$$

e.g. $\chi_3 = \frac{1}{6} = 0.167 \Rightarrow P_3 = 0.167 \times P_{\text{total}}$

Example: Partial Pressures

A gas mixture has $P_{\text{total}} = 2.0\,\text{atm}$. The mole fraction of gas 3 is $\chi_3 = \frac{1}{6}$. Find P_3.

$$P_3 = \chi_3 \times P_{\text{total}}$$

$$P_3 = \left(\frac{1}{6}\right)(2.0\,\text{atm})$$

$$\boxed{P_3 = 0.33\,\text{atm}}$$

Gas 3 is $\frac{1}{6} \approx 16.7\%$ of the mixture by moles, so it contributes 16.7% of the total pressure.

Check Your Understanding

A container holds three gases at $P_{\text{total}} = 4.0\,\text{atm}$:

$\chi_A = 0.50, \quad \chi_B = 0.30, \quad \chi_C = 0.20$

Find P_A, P_B, and P_C.

Come to lecture to see the answer.

Collecting Gas over Water

When gas is collected by water displacement, the collected gas is saturated with water vapor:

$$P_{\text{total}} = P_{\text{gas}} + P_{\text{H}_2\text{O}}$$

Pressure of the **dry** gas:

$$P_{\text{gas}} = P_{\text{total}} - P_{\text{H}_2\text{O}}$$

- $P_{\text{H}_2\text{O}}$ depends on temperature (found in tables).
- At 25 °C: $P_{\text{H}_2\text{O}} = 23.8\,\text{mmHg}$.
- At 100 °C: $P_{\text{H}_2\text{O}} = 760\,\text{mmHg}$ — this is why water boils at 100 °C at 1 atm!

Outline of Part III: Gas Mixtures

Dalton's Law and Partial Pressures

Gas Stoichiometry

Gas Stoichiometry

The ideal gas law connects moles to measurable gas quantities:

$$n = \frac{PV}{RT}$$

This lets us use a gas volume the way we use the mass of a solid in stoichiometry.

General Approach

1. Convert known gas volume → moles using $n = PV/RT$ (or use STP shortcut).
2. Apply stoichiometric mole ratios from the balanced equation.
3. Convert product moles → volume using $V = nRT/P$.

At STP shortcut: $n = V/(22.41\,\mathrm{L\,mol^{-1}})$.

Check Your Understanding

For the reaction: $N_2(g) + 3\ H_2(g) \rightarrow 2\ NH_3(g)$

All volumes measured at the same T and P.

How many liters of NH_3 are produced from 5.0 L of N_2?

Come to lecture to see the answer.

Part IV

Kinetic Molecular Theory

Outline of Part IV: Kinetic Molecular Theory

Outline of Part IV: Kinetic Molecular Theory

Effusion and Diffusion

Effusion

The escape of gas molecules through a **tiny pinhole** into a vacuum or lower-pressure region.

A helium balloon slowly deflates as He atoms effuse through microscopic pores in the latex.

Diffusion

The spreading of gas molecules through space in response to a **concentration gradient** (high $\rightarrow$ low concentration).

The scent of perfume gradually spreads across a room by diffusion.

\textbf{Effusion} \textbf{Diffusion}

Graham's Law of Effusion

Graham's Law

The rate of effusion of a gas is **inversely proportional** to the square root of its molar mass:

$$\text{rate} \propto \frac{1}{\sqrt{\mathcal{M}}}$$

For two gases A and B:

$$\frac{\text{rate}_A}{\text{rate}_B} = \sqrt{\frac{\mathcal{M}_B}{\mathcal{M}_A}}$$

Lighter gases move faster $\Rightarrow$ they effuse and diffuse more rapidly.
This is because lighter molecules must travel faster to have the same kinetic energy.

Example: Graham's Law

H_2 vs. O_2 Effusion Rate

How much faster does H_2 effuse than O_2?
($\mathcal{M}_{H_2} = 2.016\,\mathrm{g\,mol^{-1}}, \quad \mathcal{M}_{O_2} = 32.00\,\mathrm{g\,mol^{-1}}$)

$$\frac{\text{rate}_{H_2}}{\text{rate}_{O_2}} = \sqrt{\frac{\mathcal{M}_{O_2}}{\mathcal{M}_{H_2}}} = \sqrt{\frac{32.00\,\mathrm{g\,mol^{-1}}}{2.016\,\mathrm{g\,mol^{-1}}}}$$

$$= \sqrt{15.87}$$

$$\boxed{\frac{\text{rate}_{H_2}}{\text{rate}_{O_2}} \approx 3.98 \approx 4}$$

H_2 effuses ≈ 4 times faster than O_2.

Intuition: $\sqrt{32/2} = \sqrt{16} = 4$.

Check Your Understanding

Which effuses faster: He ($\mathcal{M} = 4.00\,\mathrm{g\,mol^{-1}}$) or N_2 ($\mathcal{M} = 28.0\,\mathrm{g\,mol^{-1}}$)?

By approximately what factor?

Come to lecture to see the answer.

Outline of Part IV: Kinetic Molecular Theory

Effusion and Diffusion

Kinetic Molecular Theory

The Kinetic Molecular Theory of Gases

Five Postulates of KMT

1. Gas particles are in **continuous random motion**, moving in straight lines between collisions.
2. The volume of gas particles is **negligibly small** compared to the distances between them.
3. Gas pressure arises from **collisions** of particles with the container walls.
4. All collisions are **elastic** — no kinetic energy is lost.
5. The average kinetic energy of particles is **directly proportional to the Kelvin temperature**: $\overline{\mathrm{KE}} \propto T$

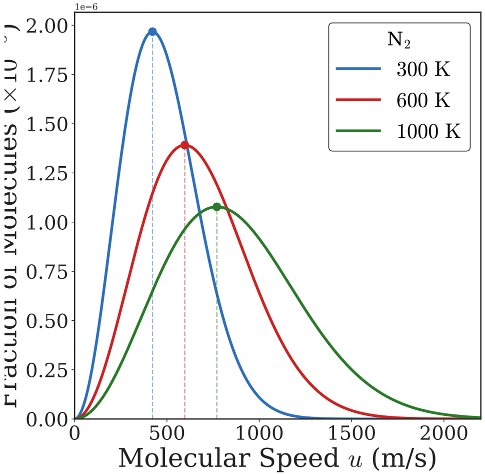
How KMT Explains the Gas Laws

Gas Law	**KMT Explanation**
Boyle's	$V\downarrow\Rightarrow$ more wall collisions per area $\Rightarrow P\uparrow$
Charles's	$T\uparrow\Rightarrow$ faster particles; expand V to keep P constant
Avogadro's	More particles $\Rightarrow$ more collisions; expand V to keep P constant
Dalton's	No intermolecular forces: each gas hits walls independently
Amontons's	$T\uparrow\Rightarrow$ harder/more frequent collisions $\Rightarrow P\uparrow$ at const V

The Maxwell-Boltzmann Speed Distribution

Gas molecules do not all travel at the same speed. The **Maxwell-Boltzmann distribution** shows the fraction of molecules at each speed.

- **Higher** T: peak shifts to higher speeds; curve broadens and flattens.
- **Higher** $\mathcal{M}$: peak shifts to *lower* speeds (heavier $\Rightarrow$ slower at same KE).



Check Your Understanding

Two gases at the same temperature: He and Xe.

Which has the higher average molecular speed? Explain why.

Come to lecture to see the answer.

Root-Mean-Square Speed

RMS Speed (u_{rms})

The most commonly used average molecular speed:

$$u_{\text{rms}} = \sqrt{\frac{3RT}{\mathcal{M}}}$$

$R = 8.314\ \text{J·mol}^{-1}\text{·K}^{-1}$; T in K; $\mathcal{M}$ in kg mol^{-1}; result in m s^{-1}.

- Use $R = 8.314\ \text{J·mol}^{-1}\text{·K}^{-1}$ (not 0.08206).
- Molar mass **must** be in kg mol^{-1} (divide g/mol by 1000).

Example: RMS Speed Calculation

RMS Speed of N_2 at $30\,°C$

Calculate u_{rms} for nitrogen gas at $30\,°C$.

$$T = 303\,K, \quad \mathcal{M}_{N_2} = 0.028\,01\,kg\,mol^{-1}$$

$$u_{rms} = \sqrt{\frac{3RT}{\mathcal{M}}}$$

$$= \sqrt{\frac{3(8.314)(303)}{0.02801}}$$

$$\boxed{u_{rms} = 519\,m\,s^{-1}}$$

- $519\,m/s \approx 1160$ mph!
- Very fast — explains rapid mixing of gases.
- Real diffusion is slower due to molecular collisions.

Check Your Understanding

If the temperature of a gas doubles from 300 K to 600 K, by what factor does u_{rms} change?

Come to lecture to see the answer.

Part V

Non-ideal Gas Behavior

Outline of Part V: Non-ideal Gas Behavior

Outline of Part V: Non-ideal Gas Behavior

Non-ideal Gas Behavior

When Does the Ideal Gas Law Break Down?

Conditions for Non-ideal Behavior

- **High pressure**: molecules are crowded together; their finite volume is no longer negligible; repulsions become significant.
- **Low temperature**: slow-moving molecules feel intermolecular attractions more strongly.

Two KMT Postulates That Fail

- Postulate 2 (negligible molecular volume) breaks down at **high** P.
- Postulate 3 (no intermolecular forces) breaks down at **low** T.

Most ideal: **low** P and **high** T

The van der Waals Equation

Modifies the ideal gas law with two correction terms:

$$\left(P + \frac{n^2 a}{V^2}\right)(V - nb) = nRT$$

Constant a

Corrects for **intermolecular attractions**.

Larger $a \Rightarrow$ stronger attractions.

The $n^2 a/V^2$ term adds to P (attractions reduce the effective pressure).

Constant b

Corrects for **finite molecular volume**.

Larger $b \Rightarrow$ larger molecules.

The nb term reduces V (molecules take up space that is unavailable to other molecules).

Van der Waals Constants

Gas	**Formula**	a ($\mathrm{L^2\,atm\,mol^{-2}}$)	b ($\mathrm{L\,mol^{-1}}$)
Helium	He	0.034	0.0237
Hydrogen	H_2	0.245	0.0265
Nitrogen	N_2	1.39	0.0391
Oxygen	O_2	1.36	0.0318
Carbon dioxide	CO_2	3.59	0.0427
Water	H_2O	5.46	0.0305

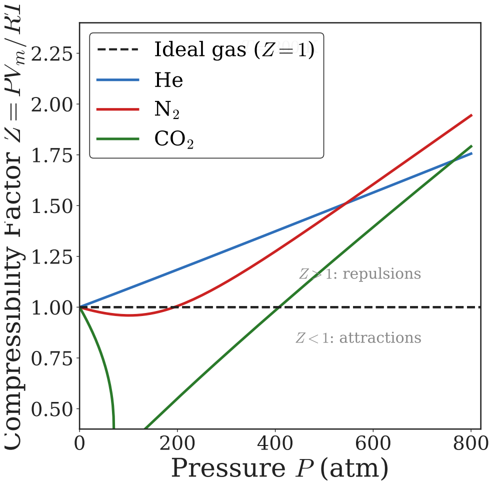
- H_2O has the largest a $\Rightarrow$ strongest intermolecular attractions (hydrogen bonding).

Real vs. Ideal Gas Behavior

Compressibility Factor Z

$$Z = \frac{PV}{nRT}$$

- Ideal gas: $Z = 1$ always.
- $Z > 1$: repulsions dominate; gas harder to compress than ideal.
- $Z < 1$: attractions dominate; gas easier to compress than ideal.



Check Your Understanding

Which gas deviates **most** from ideal behavior at high P and low T?

(A) He (B) H_2 (C) N_2 (D) CO_2

Come to lecture to see the answer.

Check Your Understanding

Under what conditions of T and P does a real gas behave most like an ideal gas?

Explain using the KMT postulates.

Come to lecture to see the answer.